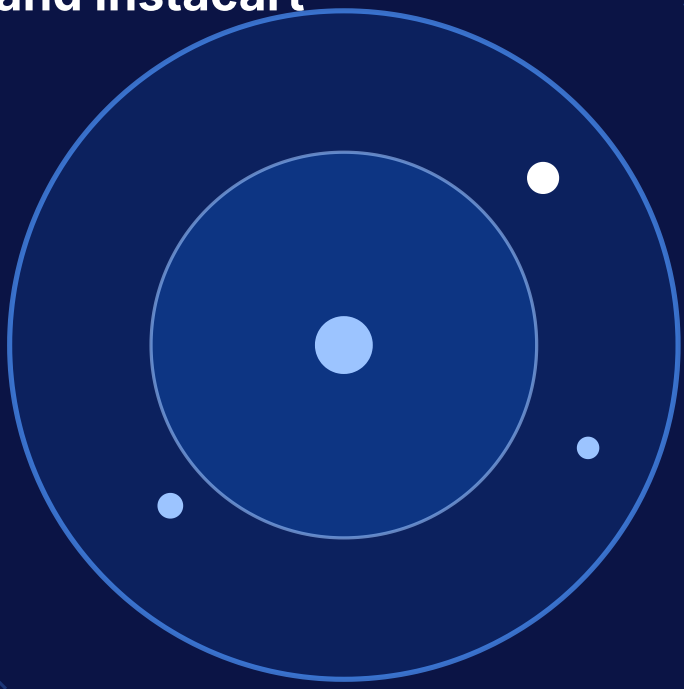
Agent-Ready Instacart

When a customer's own AI assistant shops for them, **the customer sets the rules and Instacart enforces them.**



Q1 Where to start

Q2 The board demo

Q3 Which API first

Q4 "Quantity: 2"

Q5 How much autonomy

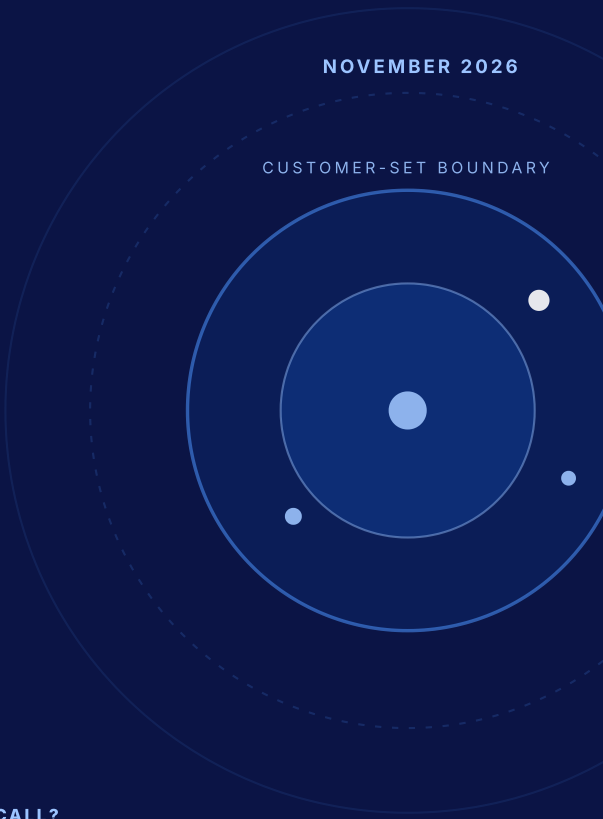
Q6 Should we worry?

Varun Kale

ED22B018 · Indian Institute of Technology Madras

Instacart case study

Six product decisions, one live prototype



Agent-Ready Instacart

A customer's own AI assistant builds the weekly grocery basket. **The customer sets the rules. Instacart enforces them.** Six product decisions, one live prototype.

THE ONE TEST BEHIND EVERY DECISION · WOULD THE CUSTOMER HAVE MADE THIS CALL?

✓ ACT

Yes → Act

Inside the customer's explicit rules. The assistant proceeds and shows why.

◇ ASK

Unsure → Ask

An unresolved choice goes back to the customer, once, with the options.

× BLOCK

No → Block

Breaks a firm rule. Instacart stops it, whatever the assistant asked.

SIX QUESTIONS · SIX CALLS · THE NUMBER BEHIND EACH

Q1 · WHERE TO START

Start with the weekly repeat basket

~52

shopping occasions a year per weekly customer

Q2 · THE BOARD DEMO

Push back, but keep the date

6 weeks

to demo a real stockout, not a shortcut

Q3 · WHICH API FIRST

Substitution preferences, not checkout

2 × 4

engineers × weeks to ship one rule check

Q4 · "QUANTITY: 2"

Our contract lost the unit; fix it once

$3.60 vs $6.40

same request, double the flour

Q5 · HOW MUCH AUTONOMY

Cap at 4 orders per rolling 30 days

≤ 0.5%

unintended orders before the cap grows

Q6 · SHOULD WE WORRY?

"It could. Here's how we'd know early."

5 gates

each with an owner and a pause trigger

EVIDENCE THE DECK BUILDS ON

840k+

Instacart customers in a field experiment on low-stock disclosure

−7.5%

poor replacement ratings when stock was disclosed in that test

1 in 5

UK online grocery shoppers got a substitution in their last order

$320

maximum automatic spend a month at the Q5 cap ($80 × 4)

LIVE PROTOTYPE · 60-SECOND AUTOPLAY OR 90-SECOND TOUR

Watch Maya's assistant act, ask and block →

inmobi-prototype.vercel.app · choose "Show me how it works" to see the request, rule and verdict behind every step.

Varun Kale

ED22B018 · IIT Madras

Ship trust first, autonomy second.

Placement submission · November 2026

EXECUTIVE SUMMARY · THE WHOLE DECK ON ONE PAGE

The customer sets the rules. Instacart enforces them.

Agents won't fail because they can't click "checkout". They fail by acting confidently outside what the customer meant. So we ship **trust first, autonomy second**.

01 THE BRIEF

Make Instacart usable by an assistant the customer brought

Within **12 months** agents will compare, order, cancel and adjust for users, often with nobody checking each step. The assistant shipped with the phone, not with Instacart.

02 THE REAL RISK

Confidently wrong, not slow

Already **1 in 5** online grocery shoppers get a substitution. An agent repeats a wrong swap or a misread "2" every week, and one viral "AI ordered the wrong groceries" story outweighs months of growth.

03 OUR ANSWER

Rules live at Instacart, every assistant reads them

Saved rules + an explicit contract + a decision for every action: ACT ASK BLOCK.
Autonomy is earned with data, never granted upfront.

04 SIX QUESTIONS, SIX CALLS

Each row is one page of evidence. Highlighted numbers are the ones to remember.

Q	THE QUESTION	OUR CALL	WHY, IN ONE LINE	WE'LL WATCH · INITIAL TARGET
Q1 p4	Why start with groceries?	**Start with weekly repeat orders.** Familiar baskets make every change visible.	**~52** shops a year, each mistake is one refundable line.	Review time vs regular reorder; repeat delegation
Q2 p5	Flashy board demo or push back?	**Push back, but don't delay.** Keep the date; demo a real stockout with customer review.	The shortcut breaks within **3 months** of real agent traffic.	All agreed demo cases pass; **≥70%** of demo work reused
Q3 p6-7	Checkout or substitution API?	**Substitution preferences.** Verify the existing customer-checkout handoff in week 1.	Judgment lives in stockouts; checkout already has a review path.	Swap acceptance **≥80%**
Q4 p8-9	Why did "quantity: 2" differ?	**Our contract lost the unit.** Require SKU + unit; preview before payment.	Same words: **$3.60** for 1 kg vs **$6.40** for 2 kg.	**0** silent mismatches in shared tests
Q5 p10-11	Orders per user per month?	**4 per rolling 30 days**, max 1 per approved weekly window, within the spend cap.	Matches a weekly habit; caps exposure at **$320**.	Unintended orders **≤0.5%**, at 95% confidence
Q6 p12	"Will this happen to us?"	**"It could. Here's how we'd know early."**	Owners, evidence and pause rights agreed before launch.	Every gate has an owner and a decision

05 THE STORY IN FIVE MOVES

Same order as the deck

Q1 · ARENA — **Pick the weekly shop** — Frequent, repeatable, cheap to fix →
Q2-3 · PRIORITISE — **Build the rules first** — Demo and sprint on what lasts →
Q4 · INTERPRET — **Make the unit explicit** — One contract for all agents →
Q5 · AUTONOMY — **Earn the right to order** — Cap, test, then grow →
Q6 · LEAD — **Say what we'd watch** — Gates with owners

WEEKS 1-4 — **Substitution rules**
WEEKS 5-6 — **Unit contract + board demo**
WEEKS 7-12 — **Customer-checkout pilot**
THEN, IF GATES PASS — **Capped-autonomy test**

06 SEE IT WORK

Live prototype

Maya's Sunday basket, end to end

A real stockout (**Act $4.50 · Ask $5.50 · Block regular milk**), an ambiguous "2", and a customer-placed order of **$70.77**, **$9.23** under her cap.

inmobi-prototype.vercel.app · 60-s autoplay

60 s autoplay, hands-free
5 stops guided tour, you tap Ask
3 layers request, rule, verdict

07 SCOPE WE COMMIT TO IN THE FIRST 90 DAYS

- **One store, packaged staples**: no loose-weight produce until the unit contract is proven.
- **Assistant prepares, customer places**: autonomous ordering is a separate, opt-in test.
- **Rules are customer-owned**: agents can read them, not edit them, in the pilot.
- **Two agent platforms** integrate against one shared, versioned contract.

08 ASSUMPTIONS

- The assistant is a **third party** (shipped with the phone) connecting through Instacart's API or MCP.
- Prices, baskets and pilot dashboards are **illustrative**; targets are proposed pilot settings.
- There is **no history** of agent-placed orders (per the brief), so Q5 creates the data first.
- External sources are listed in the Appendix; anything unsourced is reasoning.

How to read each page: the navy strip under every headline holds the call, why, the trade-off and the metric. Read only the strips and you have the whole argument.

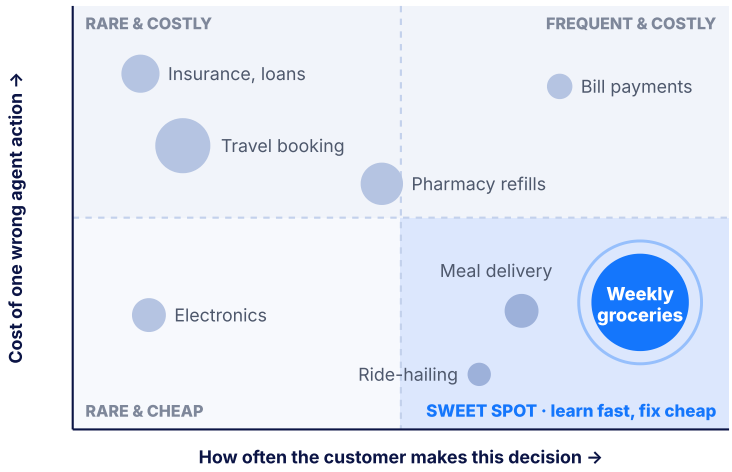
Q1 · MAKE THE CASE FOR GROCERIES

Weekly groceries let us test judgment on a basket the customer already knows.

THE CALL	WHY	TRADE-OFF ACCEPTED	WE'LL WATCH
Start with repeat customers and their usual weekly basket.	Frequent, repeatable decisions where each mistake is **1 line**, visible at the door and refundable.	Taste and freshness are subjective, so per-item rules come first (Q3).	Review time vs a regular reorder; share of customers who delegate again.

01 WHERE SHOULD AGENTS LEARN FIRST?

Directional, my judgment



Bubble size = judgment needed per order (swaps, sizes, multi-item trade-offs). Meal delivery is as frequent, but one restaurant is one decision.

WHY NOT TRAVEL
Rare, and one wrong booking is costly and hard to undo.

WHY NOT BILLS
Frequent, but little judgment: nothing to learn from.

WHY GROCERIES
Frequent, judgment-heavy and cheap to fix: refund one item, not the trip.

02 WHAT THE EVIDENCE SAYS

Sources in appendix

~52
shopping occasions a year for a weekly customer: **52 chances to learn**

1 in 5
UK online grocery shoppers got a substitution in their latest order (**29%** the year before)

INSTACART FIELD EXPERIMENT · 840,000+ CUSTOMERS

−2.9%	−7.5%	−3.7%
product replacements	poor replacement ratings	refunds

...just from **telling customers stock was low**. Stockout handling measurably moves satisfaction on Instacart itself.

03 THE ANSWER IN ONE LINE

A familiar basket makes every change visible. A stockout tests whether the assistant follows a **saved rule**, cheaply and every week.

04 FIVE REASONS THE WEEKLY BASKET WORKS

Each reason, and what it means for the pilot

1 Cadence
A similar trip every week gives another chance to see what works.
→ COMPARE WEEK ON WEEK

2 Baseline
The saved list exposes a changed item, size or price instantly.
→ REVIEW ONLY CHANGES

3 Judgment
A missing staple shows which preferences the assistant can apply.
→ TEST REAL CHOICES

4 Contained
Packaged items and customer checkout keep the first test bounded.
→ LIMIT EXPOSURE

5 Habit
Saved rules carry forward, so each order needs less setup.
→ TRACK REPEAT USE

05 FIT BY JOB IN THE BRIEF

Illustrative fit

JOB IN THE BRIEF	REPEATS	JUDGMENT	UNDO
Rebuild the weekly basket	●	◐	●
Maintain the recurring list	●	○	●
Handle a substitution	◐	●	●
Confirm and place the order	●	◐	○
Cancel or adjust after placing	◐	◐	●

● High ◐ Medium ○ Low · **Highest judgment: substitutions → build them first (Q3)**

06 WHAT CAN GO WRONG, AND WHERE THE DECK HANDLES IT

RISK	HANDLED BY
Wrong replacement · diet, taste, brand	Q3 Per-item rules; block hard mismatches
Wrong quantity · each, pack or weight	Q4 Explicit unit + checkout preview
Unwanted order · whole basket at risk	Q5 Customer checkout first, then a cap
Support cost · thin grocery margins	Q5 Refunds + contacts per order

Q2 · THE FLASHY BOARD DEMO

Push back, but don't delay. Keep the board date; change what we demo.

THE CALL	WHY	TRADE-OFF ACCEPTED	WE'LL WATCH
Keep the 6-week date. Demo a real stockout with customer review instead.	The requested shortcut is likely thrown out within **3 months** of real agent traffic. The rules demo becomes the pilot.	A narrower, less flashy story for the board.	All agreed demo cases pass; **≥70%** of demo work reused in the pilot.

01 THE ASSUMED REQUEST

Brief leaves it open

"One-tap reorder: the assistant re-buys the whole basket, no review."

Looks great on stage. Under real agent traffic it fails three ways:

1 · Stockout with no saved rule
The assistant guesses a swap. Wrong item, at the door.

2 · Price moves after the quote
The basket silently goes over the customer's cap.

3 · Agent retries a timed-out call
No order ID, so a retry places a **duplicate order**.

02 WHAT THE BOARD SEES IN WEEK 6

Controlled demo · illustrative prices

"Get my usual Sunday basket for $80 or less. If my milk is out, lactose-free up to $5 is fine."

✓ ACT	◇ ASK	× BLOCK
$4.50 lactose-free Inside the rule. Added, reason shown.	**$5.50 lactose-free** $0.50 over the limit. One-time OK.	**$3.29 regular milk** Breaks the lactose-free rule.

THEN "**Wheat ×2**" is held: 2 × 1 kg = 2 kg, **$6.40** previewed → **THEN** Cap checked: **$70.77**, $9.23 under $80 → **FINALLY** **The customer places it**, live, on stage

Every step runs in the live prototype. **The demo is the pilot's first test case**, not a throwaway.

03 THREE OPTIONS FOR THE BOARD REQUEST

TEST	BUILD AS ASKED	PUSH BACK AND DELAY	✓ KEEP DATE, CHANGE SCOPE
Week 6	Shows the original feature	Misses the board checkpoint	**Shows both swaps, the unit check and customer checkout**
Month 3	Rework once live traffic exposes the gaps	Stronger build, feedback comes later	**Same rules and cases re-run on pilot traffic**
Sponsor	Happy now, blamed later	Loses momentum and trust	**Keeps the date and a credible story**
Verdict	× Throwaway work	× Lost checkpoint	✓ Recommend

04 SIX WEEKS TO THE BOARD MEETING

WEEK 1
Verify the handoff
Prepared cart reaches customer checkout with the right items and total.

WEEKS 2-4
Build and test the rule check
Saved milk rule, Act / Ask / Block verdicts with reasons, the unit hold for "2", both swap cases on two test adapters. Same sprint as Q3.

WEEK 5
Rehearse failure
Run a failed handoff; record the fallback.

WEEK 6
Show and ask
Run the demo; ask for a customer-checkout pilot.

05 WHAT I'D TELL THE SPONSOR

"We keep the date. Let's show the board when the assistant can choose a swap and when it has to ask. Those are the same rules the pilot will run on."

06 WHO DECIDES

Instacart product lead and our **engagement lead** sign off on scope, risks and the fallback in a one-page risk note.

Agreed in week 1, so it's not a fight in week 5.

07 IF I'M OVERRULED

- Build it as a **time-boxed staging mock-up**, not production.
- Record what we expect to break.
- Protect the 4-week rules sprint; book a **month-3 review**.

Q3 · 2 ENGINEERS, 4 WEEKS: CHECKOUT API OR SUBSTITUTION-PREFERENCE API?

Substitution preferences first. That is where trust is won or lost.

THE CALL	WHY	TRADE-OFF ACCEPTED	WE'LL WATCH
Make the substitution-preference API agent-readable. Verify checkout in week 1.	Judgment lives in stockouts. Checkout already has a documented customer review path.	Assistants check out clumsily (via a handoff) for one more cycle. Clumsy is safe.	Swap acceptance **≥80%** of proposed swaps kept at checkout.

01 SAME 8 ENGINEER-WEEKS, TWO WAYS TO SPEND THEM

CRITERION	SUBSTITUTION PREFS	CHECKOUT
Handles the hardest moment (a stockout)	●●●	●○○
Keeps the customer's preferences	●●●	○○○
New capability vs what exists today	●●◐	●○○
Risk if left unready	Wrong items, confidently	Slower, but reviewed
Smallest useful build	Read 1 saved rule, judge 1 candidate	Quote + link to review screen
What we learn	Do customers keep proposed swaps?	Does a basket reach checkout intact?

Page 3's fit check scored substitutions highest on judgment. More filled dots = better fit.

COST OF BEING WRONG · A REGRET CHECK

IF WE PICK...	...AND THE TRUTH IS	WE LOSE
Substitutions	Checkout handoff is broken	**1 week**: the week-1 gate catches it, we switch
Checkout	Checkout handoff already works	**8 engineer-weeks** rebuilding a path that exists

Substitutions-first has the smaller worst case. The gate makes it a **reversible** decision.

02 WHAT ALREADY EXISTS

CHECKOUT
Handoff to customer review
Instacart documents third-party AI apps sending a cart to customer checkout.

PREFERENCES
Replacement instructions
Storefronts already let customers save how items should be replaced.

So: checkout needs verifying, not rebuilding. Preferences exist but agents can't read or apply them yet.

MISSING FOR AGENTS TODAY

- A machine-readable rule an outside assistant can read
- A verdict with a reason it can relay to the customer
- A version number, so a stale rule is never applied

03 WHY JUDGMENT COMES FIRST

Stockouts are where the assistant must act like a **trusted personal shopper**, the brief's own words. In Instacart's own experiment, better stockout handling cut poor replacement ratings by **7.5%**.

−2.9%	−7.5%	−3.7%
replacements	poor ratings	refunds

A faster checkout without the customer's limits just ships wrong orders faster: the exact complaint Q5 tests for.

04 STEELMAN: "CHECKOUT IS THE DEPENDENCY"

The case for checkout: a broken handoff blocks the whole pilot, and a readable quote (fees, totals, requirements) helps every agent.

Why we still defer it: customers can already finish checkout in Instacart. Only the judgment step is missing. We test the handoff in week 1, so the dependency risk is retired in **5 days**, not assumed.

05 WEEK-1 RISK GATE

Decide on evidence, not opinion

Open the prepared cart as the customer. Check store, exact quantities, fees and total. Place one customer-confirmed test order.

✓ HANDOFF PASSES
Build substitutions
Reuse existing checkout. Confirm access to saved rules and catalog.

× HANDOFF FAILS
Switch the sprint to checkout
Build a reviewable quote and a checkout link first.

INITIAL TARGET
80%+
swap acceptance

06 HOW WE MEASURE SUCCESS

Swap acceptance
Share of proposed swaps shown at review that the customer keeps at checkout.

Edits per stockout
How often the customer has to fix a swap or answer a prompt.

Review time
Seconds to approve the basket vs a regular manual reorder.

Q3 · THE FOUR-WEEK BUILD

Two engineers can ship one rule check in four weeks.

THE SCOPE
One store, packaged staples, rules the customer has already saved.

WHY THIS SMALL
One verdict API, used by every agent, beats four half-built features.

DEFERRED
Agents editing rules, new payment work, swaps after checkout.

RELEASE GATE
Both adapters pass **100%** of scoped cases. Any rule-breaking swap blocks rollout.

01 THE PROPOSED API, IN ONE CALL

Illustrative

The assistant asks: **"Can I use this replacement?"** Instacart answers Act, Ask or Block, with a reason.

REQUEST · ASSISTANT → INSTACART

```
{"cart_id":"draft_17", "line_id":"milk_01",
 "candidate_sku":"lf_1l"}
```

CUSTOMER'S SAVED RULE · READ BY INSTACART

```
{"requires":"lactose_free", "pack_ml":1000,
 "ask_above_cents":500, "rule_v":3}
```

VERDICT ON THE $4.50 CANDIDATE

```
{"decision":"ACT", "reason":"WITHIN_RULE", "rule_v":3}
```

EDGE RESPONSES THE ASSISTANT MUST HANDLE

```
{"decision":"ASK",   "reason":"RULE_MISSING"}
{"decision":"BLOCK", "reason":"RULE_CONFLICT"}
{"decision":"ASK",   "reason":"STALE_RULE_VERSION"}
```

Instacart re-checks live price and stock for that SKU at cart update and at checkout.

02 HOW INSTACART DECIDES

Same rule, any agent

The usual milk is out. Each candidate is checked against the customer's saved rule:

✓ ACT	**Lactose-free, 1 L, $4.50** Inside the rule. Added to the draft with the reason.	**$4.50**
◇ ASK	**$5.50, or no rule saved** Request one-time approval, or skip. The rule stays the same.	**$5.50**
× BLOCK	**Regular milk** Conflicts with a firm rule. Omitted, and the assistant is told why.	**$3.29**

Data changed or missing? Re-check at cart update and at checkout. Still missing → ask for a verified option or skip.

ONE ENDPOINT, THREE USERS · WHY THIS IS A PRODUCT, NOT A DEMO

Outside assistants	Instacart app	In-store shoppers
Propose swaps before checkout	Explain swaps to the customer	Same rule when picking

03 TWO ENGINEERING LANES, FOUR WEEKS

8 engineer-weeks in total

	WEEK 1	WEEK 2	WEEK 3	WEEK 4 · SHIP
Engineer A Rules + API	Map saved rules and the 3 outcomes	Expose rule read + candidate check	Re-check price, stock, rule version	Stage behind a flag; verify hard blocks
Engineer B Cart + adapters	**Verify checkout handoff** (the week-1 gate)	Connect draft changes; log every verdict	Wire both test adapters to the same API	Replay shared cases; fix platform drift

04 WHAT BOTH ADAPTERS MUST PASS

TEST	PASS CONDITION
Same verdict	Same candidate + same rule → same decision and reason, on both platforms
Stale quote	A changed price or rule triggers a fresh check before the cart updates
Permissions	Unlinked or wrong-account calls fail without exposing the rules
Customer review	Final cart shows every swap; the customer confirms checkout

05 FIRST PILOT SCOPE

Ship: saved item rules, read access for the linked assistant, swap checks on the draft cart.

Defer: agents editing rules, new payment work, swaps after checkout. Instacart shoppers keep using the existing in-store replacement flow.

06 DECISION LOG

Every swap records **adapter, candidate SKU, rule version, decision, reason**, so support can trace any wrong swap in one look. This log also powers Q6.

Q4 · "QUANTITY: 2" READ TWO WAYS: WHERE IS THE PROBLEM?

The bug most likely lives in our contract: "quantity" carries no unit.

THE CALL	WHY	IMMEDIATE RESPONSE	WE'LL WATCH
Treat it as a contract bug: fix once at the source, not in each agent.	Two platforms reading one field two ways means the field is underspecified. Products sell per item, per pack and by weight.	Hold the ambiguous line; ask what "2" counts before the cart updates.	**0** silent mismatches in shared tests.

01 SAME INPUT, TWO CARTS

Illustrative variants and prices

```
{"store_sku":"wheat_flour", "quantity":2}
```

PLATFORM A READS
2 × 500 g packs
1 kg
$1.80 each → **$3.60**

PLATFORM B READS
2 × 1 kg packs
2 kg
$3.20 each → **$6.40**

Same words: **2×** the flour and **+$2.80** on one line. Neither agent "misbehaved": each filled a gap we left open.

WHY ONE NUMBER CAN'T WORK · HOW GROCERIES ARE ACTUALLY SOLD

ITEM	SOLD BY	WHAT "2" COULD MEAN
Wheat flour	Pack (500 g or 1 kg)	2 packs, but of which size?
Bananas	Each or bunch	2 bananas or 2 bunches
Tomatoes	Weight (lb)	2 tomatoes or 2 lb

02 IMMEDIATE RESPONSE

Hold the ambiguous line

Ask what "2" counts and show the store's available variants before updating the cart.

If an order was already placed

Find it by trace ID, check what was bought, correct confirmed errors, refund the difference.

03 LEADING HYPOTHESIS

The MCP tool description and the API schema left the unit implicit, so each integration inferred one. **The trace confirms which layer** before we assign the cause.

- We publish the contract, so **we own the ambiguity**.
- Fixing it in one agent leaves every future agent exposed.
- Blaming partners slows the fix and hurts the platform.

04 SIX PLACES THE UNIT CAN BE LOST

From the customer's words to the cart

MOST LIKELY · FIX HERE

CHECK **Customer's words** — What was "two" meant to count? ›
CHECK **Assistant** — Compare the selected SKU and raw arguments. ›
MOST LIKELY **MCP tool** — Does the tool description define the unit? ›
MOST LIKELY **API schema** — Does a default or mapping drop the unit? ›
CHECK **Catalog** — Sold by each, pack or weight for this SKU? ›
CHECK **Cart** — Stored line vs preview vs placed order.

05 FIND THE FIRST DIVERGENCE

A 3-question decision tree

IF BOTH AGENTS PRODUCED...	...THEN THE FAULT IS IN
the **same saved line**, different labels	**Display** preview or receipt wording
the **same raw request**, different validated lines	**Mapping** catalog or contract
the **same validated line**, different orders	**Checkout** cart version or re-quote
different raw requests	**Contract** the field let them differ

06 CONTROLLED REPLAY

1. **Pin the inputs.** Same wording, store, starting cart and catalog snapshot.
2. **Trace both paths.** Join raw call → validated line → preview → order by trace ID.
3. **Isolate the layer.** Replay through each adapter, then call the API directly. Change one layer at a time.

Q4 · HOW WE FIX IT

One shared, versioned contract makes the unit explicit for every agent.

THE FIX	WHY IT HOLDS	TRADE-OFF ACCEPTED	WE'LL WATCH
Require the exact SKU and sale unit. Preview the line before payment.	Prevent, detect and recover in three layers, so no single check has to be perfect.	One extra prompt when an agent omits the unit.	Quantity corrections per **1,000** previewed carts; refunds after payment.

01 FROM AMBIGUOUS TO EXPLICIT

Illustrative

BEFORE · V1

```
{"quantity":2}
```

AFTER · V2 REQUEST

```
{"store_sku":"wheat_pack_1kg",
 "quantity":{"value":2,"unit":"pack"}}
```

CATALOG FOR THAT VARIANT

```
{"sold_by":"pack", "net_g_per_pack":1000}
```

ERROR WHEN THE UNIT IS MISSING

```
{"error":"UNIT_REQUIRED", "allowed":["1 kg pack","500 g pack"]}
```

RESULT ON BOTH PLATFORMS

```
{"line":"2 × 1 kg pack", "total_g":2000, "subtotal":6.40}
```

Same input, same line. The unit travels with the number, so the assistant cannot guess it and the cart cannot reinterpret it.

The assistant relays one plain question: **"1 kg packs or 500 g packs?"**

02 CUSTOMER CHECKOUT PREVIEW

What Maya sees

Wheat flour, 1 kg pack

2 packs — 2 kg total

Price per pack	$3.20
Line subtotal	**$6.40**

Both agents display this **validated** line. The customer confirms in Instacart; the amount can't silently change after this.

03 FIX THE AGENT-FACING TOOL DESCRIPTION TOO

MCP tool schema

```
"quantity": { "type":"object", "required":
["value","unit"],
  "unit": { "enum":["each","pack","g"] },
  "description":"Count of the exact store variant.
   Never infer a unit; ask the customer." }
```

Agents read the description to decide what to send. Saying it plainly removes the guess at the source.

04 WHERE INSTACART ENFORCES IT

01 PREVENT
One versioned schema for API and MCP. Exact SKU, whole packs, grams for loose weight.

02 DETECT
Validate unit and value against the catalog. **Flag >3×** the usual quantity.

03 RECOVER
Bind the preview to the cart version. **Re-quote changes** before payment.

05 CONTRACT TESTS BOTH PLATFORMS MUST PASS

CASE	EXPECTED RESULT
Valid 2-pack order	Saved line and preview agree: SKU, 2 packs, 2 kg, $6.40
Missing unit / bad count	UNIT_REQUIRED; the cart stays intact
Weight on a pack SKU	Rejected; the valid pack option is returned
Quantity spike	7 packs vs a usual 2 → customer review (>3×)
Change after preview	Re-quote; the customer reviews the changed line

06 MIGRATION AND ROLLBACK

01 VERSION Require v2 for new agent writes; keep v1 only where the unit is already explicit.

02 REPLAY Compare both adapters on sample traffic with writes switched off.

03 ENABLE One adapter at a time. Pause writes on any confirmed silent mismatch.

RELEASE BAR · BEFORE PAYMENT

0 silent mismatches in shared tests. Track quantity corrections per 1,000 previewed agent carts.

AFTER PAYMENT

per 1,000 quantity-related refunds per 1,000 agent orders; every confirmed mismatch traced to its cart and schema version.

Q5 · THRESHOLD OF AGENT ORDERS PER USER PER MONTH, WITH NO HISTORY

Cap autonomous orders at four per rolling 30 days.

THE CALL
4 orders / rolling 30 days, max 1 per approved weekly window, inside the spend cap.

WHY FOUR
Matches a weekly habit (**4.3** weeks a month) and caps exposure at **$320** at an $80 basket.

TRADE-OFF ACCEPTED
The 5th and any off-schedule order go back to customer checkout.

WE'LL WATCH
Unintended orders **≤0.5%**, proven on p11.

01 THE STARTING LIMIT

4 automatic orders per account, rolling 30 days

- **≤1 per weekly window** the customer approved (e.g. Sunday 10-12)
- **Inside the spend cap**, all-in with fees and tax
- **Only after 3 confirmed baskets** and explicit opt-in
- **Revocable** in one tap, any time

WHY ROLLING 30 DAYS, NOT PER CALENDAR MONTH
A calendar cap of 4 allows **8 orders in 8 days**: four on Jan 25-31, four more on Feb 1-7. A rolling window never allows more than 4 in any 30 days.

WORST CASE A MONTH
$80 × 4 = $320

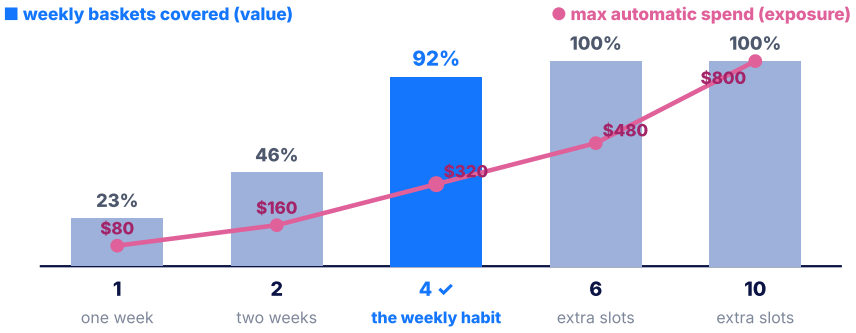
02 WHAT 30 DAYS LOOK LIKE FOR A WEEKLY CUSTOMER

Day 1	Day 8	Day 15	Day 22	Day 29
AUTO 1	AUTO 2	AUTO 3	AUTO 4	ASK · CONFIRM

A 30-day window holds about 4.3 weeks, so a 5th weekly basket can fall inside it. That one goes to customer checkout for a fresh confirmation.

03 WHY FOUR, NOT ONE OR TEN

Illustrative, $80 per order



4 is the knee: it covers **92%** of a weekly customer's baskets. Going to 6 or 10 adds almost no value for a weekly customer but raises the worst case to $480-$800, before we have any evidence.

04 QUALIFY

Confirm the first three

The customer confirms 3 assistant-built baskets and can fix the list and saved rules first.

05 AUTHORISE

Opt in explicitly

Approve the list, store, address, weekly window and an all-in spend cap. Revocable any time.

06 STAY IN CONTROL

A receipt every time

Explain what was ordered and why; offer pause and a 30-minute cancel while still eligible.

07 HOW INSTACART DECIDES AT CHECKOUT

VERDICT	WHEN IT APPLIES	WHAT INSTACART DOES
✓ ACT	Valid cart and permission, unused weekly window, fewer than 4 orders in 30 days	Re-check the all-in quote, place the order, send the receipt
◇ ASK	2nd order in a window, 5th in 30 days, off-schedule, or over the cap	Route to customer checkout; approval covers this order only
× BLOCK	Permission missing or revoked, a firm rule fails, quantity still ambiguous	Stop until the issue is resolved

08 WHAT USES A SLOT

- Orders the assistant placed **without fresh confirmation**, including ones later cancelled.
- Customer-confirmed and failed attempts use **zero** slots.
- Retries share one order ID, so a retry can't spend two slots or buy twice.
- Slots are counted across all linked assistants.

Q5 · NO HISTORICAL DATA, SO WE CREATE IT

Test the cap against customer confirmation, and only grow it with evidence.

THE CALL	WHY	TRADE-OFF ACCEPTED	WE'LL WATCH
Run a randomized pilot: customer confirms vs assistant places up to 4.	The brief says there is no history of agent orders. A controlled test gives us the first honest baseline.	Slower expansion; we need about **600** clean orders before raising the cap.	Intended orders up, review effort down, harm **≤0.5%**.

01 PILOT DESIGN · SAME BASKET, DIFFERENT CHECKOUT PERMISSION

≥30 days per account

ELIGIBLE ACCOUNTS
3 confirmed baskets + explicit opt-in
Same lists, rules and weekly windows in both groups.

⇉

CONTROL · 50%
Customer confirms each checkout
The assistant prepares; the customer places every order.

TREATMENT · 50%
Assistant places up to 4
Eligible orders inside the cap; exceptions go to the customer.

Before launch: pass tests for a 5th order, retries, revoked permission and wrong units. Set the minimum improvement and feedback coverage in advance.

02 WHAT WE MEASURE

SIGNAL	MEASURE (PER ASSIGNED ACCOUNT, PER 30 DAYS)	WHAT IT TELLS US
Value	Completed, **intended** orders, including customer checkout	Does the policy add orders people wanted?
Effort	Review minutes and confirmation prompts per scheduled basket	Do customers supervise less?
Trust	Confirmed unintended orders ÷ autonomous orders; refunds and support contacts per order	Does saved time cost more wrong orders?

HOW WE VERIFY INTENT

"Did you mean for this order to be placed?"
Asked on every receipt in both groups. A "no", a cancellation in 30 minutes or a refund marked unintended counts. We audit how missing answers could hide harm.

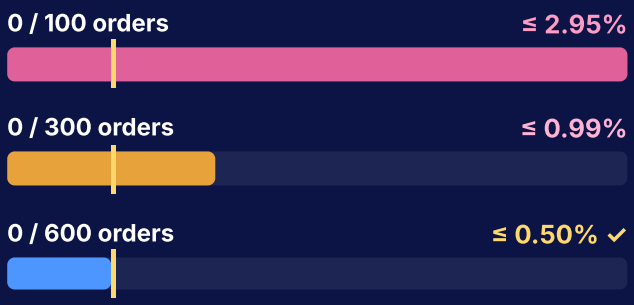
GUARDRAILS THAT CAN STOP THE TEST ON THEIR OWN

- **Opt-outs** — Customers revoking autonomy
- **Refunds** — Per order, vs control
- **Support contacts** — Per order, vs control

03 HOW MUCH EVIDENCE IS ENOUGH?

| = 0.5% ceiling · 95% confidence

With **zero** unintended orders, the true rate is at most about **3 ÷ orders** (the "rule of three").



So we need about **600** independent, intent-checked orders with no harm to show the rate is below 0.5%.

One-sided bound: we only need to show harm is **below** the ceiling, not estimate it exactly. Any verified harmful order resets the count and triggers a review.

SIZING THE PILOT

At ~4 orders per account a month, 600 orders ≈ **150 account-months**: 150 treatment accounts for 1 month, or 75 for 2. One store is enough.

04 DECISION AT EACH PILOT REVIEW

Pre-agreed, so nobody argues after the data arrives

PAUSE
Stop the affected automation
A verified order outside permission pauses it. A reported unwanted order returns that customer to confirmation while we investigate.

HOLD
Keep the cap at 4
Evidence is thin, or neither value nor effort improved. Customers keep confirming.

EXPAND
Grow at four first
Value or effort improves without the other worsening, support is stable, and the harm bound is ≤0.5%. Only then test a 5th order.

Q6 · "SHOULD WE BE WORRIED THIS WILL HAPPEN TO US?"

It could. Here's how we'd know early, and what we'd do.

THE CALL	WHY	TRADE-OFF ACCEPTED	WE'LL WATCH
Be honest about the risk; agree evidence, owners and pause rights before launch.	False reassurance is how projects lose trust. Early, specific signals keep it.	A gated, slower rollout with visible stop points.	Every gate has an owner, evidence and a decision, every **4 weeks** or sooner.

01 WHAT I'D SAY TO THE CLIENT, WORD FOR WORD

"It could, and I'd take it seriously. We don't yet know why their project stalled, and our checkout handoff and quantity checks still have to pass. I'll bring you the evidence every week and flag any blocker the moment it threatens the pilot. If a gate fails, we change scope or pause that path before more customers use it. And if we stall, we say so together, early."

Honest · specific · shared ownership · no blame on the other consulting team.

WHAT I WOULD NOT SAY

- ✕ **"It won't happen to us."** A promise nobody can keep.
- ✕ **"Their consultants were weak."** Unverified, and it erodes trust in all of us.
- ✕ **"Let's go slower to be safe."** Vague. Gates are specific.

02 AGREE TODAY: WHO CAN ACT

Instacart product lead owns scope and launch decisions.

Engagement lead (me) owns the weekly risk review and the decision record.

Engineering owner can pause an affected automation immediately, without a meeting.

03 WHY STALLS HAPPEN, AND OUR GUARD

COMMON CAUSE	OUR STRUCTURAL GUARD
Big-bang scope	One store, staples, one API
Autonomy before trust	Customer checkout first (p10-11)
No stop criteria	Pause triggers agreed upfront
Unclear owner	A named owner per gate
Hidden data gaps	Week-1 catalog audit
Public overpromise	No launch claims before p11 evidence

04 FIVE CHECKPOINTS THAT FORCE A DECISION

WHEN	OWNER	EVIDENCE TO PASS	IF IT FAILS
Week 1	Cart engineer	A prepared basket reaches customer checkout with the right items, quantities and total	Switch the sprint to checkout (p6)
Week 4	API engineer	Both adapters agree on every substitution verdict; every hard-rule violation is blocked	Fix failing cases before they appear in the demo
Week 6	Product lead	Both demo cases pass; the unit contract and preview agree across platforms	Hold the customer-checkout pilot
Before autonomy	Engineering owner	3 confirmed baskets + opt-in per account; caps, retries and revocation pass replay	Stay at customer checkout
30+ days in	Product lead	Intended orders or effort improve; support stable; harm bound ≤0.5%	Hold the cap at 4 (p11)

05 IF AN ORDER GOES WRONG · CONTAIN

Pause the affected path

A verified out-of-permission order or a silent quantity mismatch triggers a pause; orders route back to customer confirmation.

06 RECOVER

Help affected customers

Find the orders through the decision log (p7), correct confirmed errors, share one factual update with the sponsor.

07 RESTART

Require a fresh decision

Replay the failure and the fix; the product lead approves a restart only once the checks pass.

Learn from the competitor without gossip: use verified details of their setback to stress-test our own assumptions, and escalate any confirmed overlap the same week.

LIVE PROTOTYPE · BASKET, AN AGENT-READY GROCERY CONCEPT

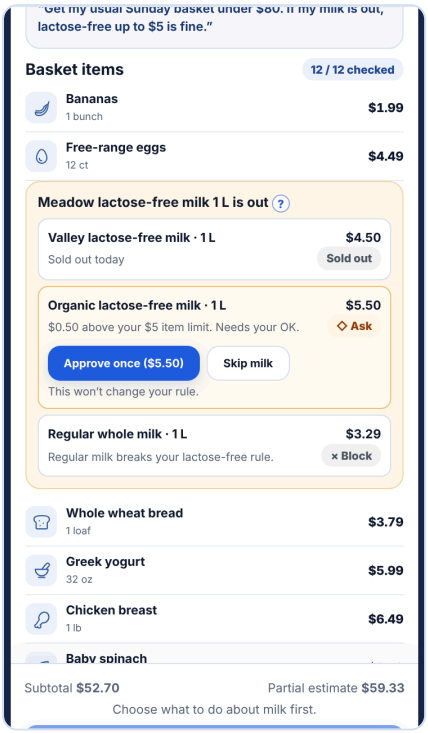
From a saved rule to an order the customer meant.

TRY IT
inmobi-prototype.vercel.app
Pick **"Show me how it works"** to see the request, rule and verdict behind every step.

60 SECONDS
▶ **Autoplay** runs the whole Sunday basket by itself.

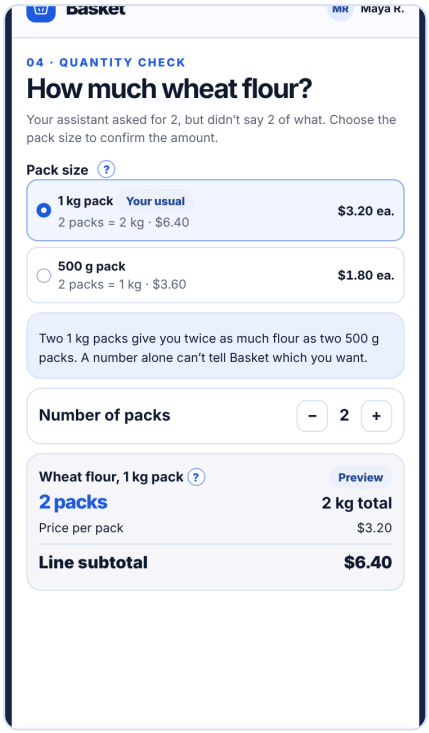
90 SECONDS
Guided tour, 5 stops, one per case question. You tap the Ask yourself.

01 Q3 ACT, ASK OR BLOCK



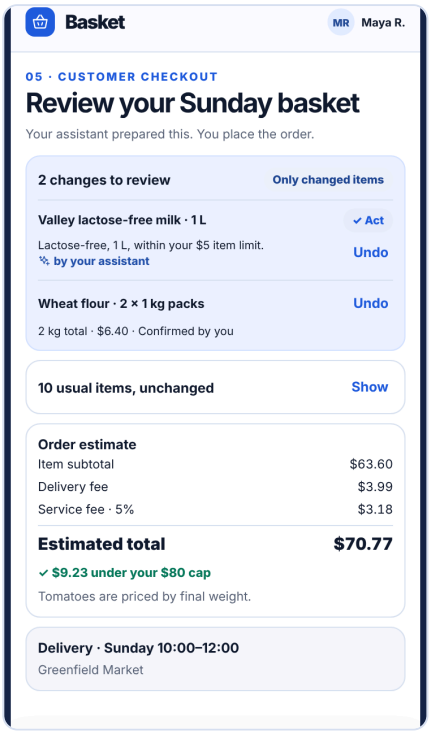
The $4.50 milk sold out, so **$5.50 waits for Maya's OK**; regular milk is blocked. Her saved $5 rule stays the same.

02 Q4 NO UNIT, NO GUESSING



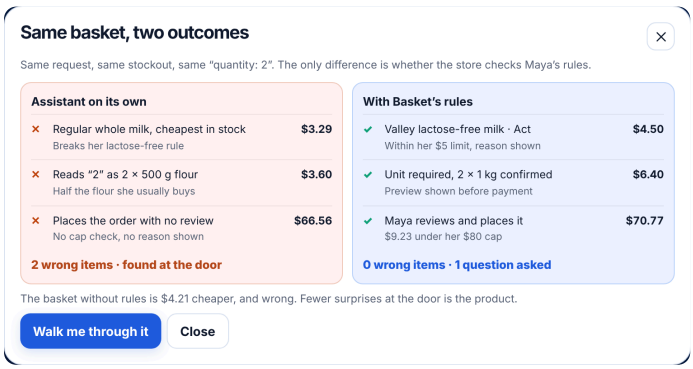
"2" is held until Maya picks the pack. The preview shows **2 × 1 kg = 2 kg, $6.40** before anything is paid.

03 Q5 THE CUSTOMER PLACES IT

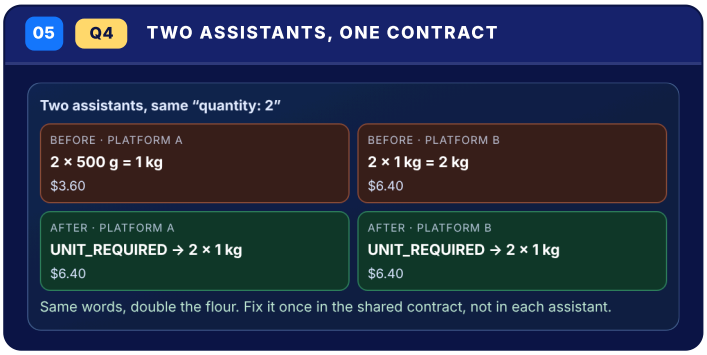


Only the 2 changes need review. **$70.77, $9.23 under her $80 cap**; "I didn't mean this" feeds the Q5 harm metric.

04 THE 5-SECOND VERSION: WITHOUT RULES VS WITH BASKET



05 Q4 TWO ASSISTANTS, ONE CONTRACT



06 ALSO IN THE PROTOTYPE

- **What-if switches:** milk sells out · assistant asks for 7 packs (3× guard) · cap set to $65
- **Scoped permissions:** the assistant can build and swap, never pay (pilot)
- **Week-6 board demo** steps are tagged in the side panel Q2
- **Activity trace + cancel** for support and incident review Q6

CLOSE · THE 90-DAY PLAN

Ship trust first, autonomy second.

Every phase has a gate. We only move on when the evidence says the customer would have made the same call.

01 ROADMAP WITH THE GATE TO THE NEXT PHASE

1	2	3	4
WEEKS 1-4	WEEKS 5-6	WEEKS 7-12	IF GATES PASS
Substitution rules	**Unit contract + board demo**	**Customer-checkout pilot**	**Capped-autonomy test**
MAYA EXPERIENCES Swaps arrive **with a reason**; regular milk never shows up.	MAYA EXPERIENCES Asked **"which pack?"** once, then it's remembered.	MAYA EXPERIENCES Reviews **only the changes**, then places the order herself.	MAYA EXPERIENCES Her opted-in basket **arrives on schedule**, with a receipt.
BUILD Saved rules, Act / Ask / Block verdict API, decision log; checkout handoff verified in week 1. GATE Both adapters agree; every hard-rule violation blocked	BUILD v2 quantity contract (SKU + unit), checkout preview, contract tests; the week-6 board demo. GATE Both platforms preview the same line; all demo cases pass	LEARN Repeat customers, one store, packaged staples. The assistant prepares; the customer places. GATE Swap acceptance **≥80%**; review faster than reorder	TEST Randomized 1:1 after 3 confirmed baskets and opt-in. Up to 4 orders per rolling 30 days. GATE Value or effort up; harm bound **≤0.5%** after ~600 orders

02 DAY-90 SCORECARD · WHAT "WORKING" LOOKS LIKE

Proposed targets

≥80%	0	< reorder	2	≤0.5%
swap acceptance: proposed swaps kept at checkout	**silent quantity mismatches** in shared contract tests	**review time** vs a manual weekly reorder	**agent platforms** live on one v2 contract	**unintended orders** before any cap above 4
Q3	Q4	Q1	Q4	Q5

03 WHAT I'D NEED FROM INSTACART

- **Catalog access**: stock, pack sizes and ingredient data for one pilot store.
- **A pilot store** with a working customer-checkout path.
- **A named product owner** who can pause, approve or reject a rollout.
- **Two agent platforms** willing to integrate against the v2 contract.

04 WHAT I'D VALIDATE FIRST

- Does the checkout handoff preserve exact lines and the all-in price?
- How often are pack and dietary data missing in the catalog?
- Will customers opt in, and reliably report intent on the receipt?
- What does a "good" swap look like by category (dairy vs produce)?

05 THE SIX CALLS, IN ONE BREATH

Q1	Q2	Q3	Q4	Q5	Q6
Weekly baskets first	**Keep the date, change the demo**	**Substitution rules before checkout**	**Make the unit explicit**	**Cap at 4 / 30 days**	**Gates with owners**

THANK YOU

See the whole argument run in 60 seconds →

inmobi-prototype.vercel.app · Varun Kale · ED22B018 · IIT Madras

Appendix

Sources and notes on every figure →

APPENDIX · SOURCES

Every external number, one click from its source.

Tap any title to open the source in your browser. Tap a page number to jump to where it is used.

01 EXTERNAL SOURCES

All links accessed November 2026

#	SOURCE	WHAT IT SUPPORTS	USED ON
1	**Which?, "The silliest supermarket substitutions revealed"** 12 Mar 2026 · Nov 2025 survey of 3,552 UK adults, 1,417 online shops	**1 in 5** online grocery shoppers got a substitution in their latest shop	p2 · p3 · p4
2	**Knight & Mitrofanov, "Disclosing Low Product Availability"** *Management Science* (2025) · Instacart field experiment	**840,000+** customers; −2.9% replacements, **−7.5%** poor ratings, −3.7% refunds	p2 · p4 · p6
3	**Instacart Help Center, "Shop with third-party AI apps"** How outside assistants hand a cart to Instacart checkout	Checkout handoff already exists, so Q3 builds substitutions first	p6
4	**Instacart Docs, "Replacement instructions"** Customers save replacement choices that carry over to future orders	Saved preferences exist but agents cannot read or apply them yet	p6
5	**Instacart Docs, "Replacements flow"** Instacart Connect post-checkout guide	Where an Act / Ask / Block verdict plugs into today's flow	p7
6	**Instacart Docs, "Product quantity"** Items sold by each, by package or by weight	Why "quantity: 2" needs a unit (each, pack, kg)	p8 · p9
7	**Model Context Protocol, "Tools" specification** Version 2025-11-25 · tool names, descriptions, input schemas	The tool description is part of the contract agents read	p9
8	**NIST/SEMATECH Dataplot, "Exact binomial"** One-sided exact upper confidence limit for a proportion	**0 / 600** orders → true rate ≤ 0.5% at 95%	p11
9	**Live prototype, inmobi-prototype.vercel.app** Built for this case · not affiliated with Instacart	Every flow in the deck, clickable	p13

02 HOW THE DECK'S NUMBERS ARE WORKED OUT

FIGURE	WORKING	PAGE
~52 occasions	One basket a week × 52 weeks	p4
$3.60 vs $6.40	2 × 500 g at $1.80 vs 2 × 1 kg at $3.20	p8
92% coverage	min(4, 4.35) ÷ 4.35 weekly baskets in 30 days	p10
$320 worst case	$80 cap × 4 orders per rolling 30 days	p10
≤ 0.5% at 600	$1 - 0.05^{1/n} \approx 3 \div n$, zero events in n orders	p11
≤ 2.95% · ≤ 0.99%	Same bound at 100 and 300 clean orders	p11
2 × 4	Two engineers × four weeks, from the brief	p7
$70.77, $9.23 under	Prototype basket total vs the $80 cap	p13
150 account-months	600 orders ÷ ~4 orders per account a month	p11

03 WHAT IS ILLUSTRATIVE, AND WHAT COMES FROM THE BRIEF

- **From the case brief:** the 12-month agent horizon, the six-week board date, the three-month concern, two engineers for four weeks, and no history of agent-placed orders.
- **Illustrative:** Maya's basket, product names, prices, the $80 cap, API fields and every dashboard figure in the prototype.
- **Proposed pilot settings, not results:** 80% swap acceptance, 70% demo reuse, the 0.5% harm ceiling, the 3× quantity flag and the four-order limit.
- **My judgment:** the arena map on p4 is directional. Checkpoint owners on p12 are roles, not named people.
- **Directional evidence:** the Which? data is from the UK; the US pilot measures its own baseline before any target is fixed.

04 SEE IT LIVE

inmobi-prototype.vercel.app

Every flow in this deck, clickable. Built for this case; not affiliated with Instacart.

← Back to the agenda